The Effect of Different Surgical Specialties on the Outcomes of Patients with Necrotizing Fasciitis: Morbidity and Mortality

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Introduction

Necrotizing fasciitis (NF) is a rapidly progressing soft tissue infection with high morbidity and mortality. Prompt surgical intervention is critical for survival. However, the impact of different surgical specialties on patient outcomes remains unclear. This study aims to evaluate the effect of surgical specialty (orthopedic vs. general surgery) on NF-related morbidity and mortality, identifying factors that influence patient survival.

Methods

This retrospective cohort study at King Abdulaziz University Hospital analyzed 71 adult necrotizing fasciitis patients treated surgically by either general surgery or orthopedic teams, comparing demographics, comorbidities, surgical details, and outcomes, while excluding those without surgery, or incomplete records.

Results and Discussion

Variable	General Surgery (%)	Orthopedic Surgery (%)	p-value
Mortality Rate	34.4	59.0	0.039
Septic Shock	15.6	51.3	0.002
ICU Admission	73.3	89.7	0.075
Median Length of Stay (days)	30.0 (IQR: 14.5-73.0)	43.0 (IQR: 22.0-86.0)	0.320
Readmission Rate	6.3	12.8	0.446

Table 1: Mortality, ICU Admission & Hospital Stay

Complication	General Surgery (%)	Orthopedic Surgery (%)	p-value
Cardiac Arrest	6.3	7.7	>0.999
Disseminated Intravascular Coagulation (DIC)	6.3	2.6	0.585
Metabolic Acidosis	3.1	5.1	>0.999
Secondary Infection (Osteomyelitis)	0.0	2.6	>0.999
Secondary Infection (Wound)	3.1	0.0	0.451
Bed Sores	3.1	0.0	0.451

Table 2: Morbidity & Complications

- **Diabetes, Hypertension & Outcomes:**
  Diabetes was present in 87.3% of patients but was not directly associated with higher mortality ($p > 0.999$). However, among diabetic patients, mortality was higher in the orthopedic group (55.9%) than in the general surgery group (35.7%). Similarly, hypertensive patients had worse survival in the orthopedic group (62.5% vs. 36.8%) despite similar mean HbA1c levels ($p = 0.468$), emphasizing the need for aggressive comorbidity management.

- **Amputation Rates & Surgical Outcomes**

Amputation was performed in 47.1% of cases, with above-knee amputations accounting for 60.6% of these. The orthopedic group had a higher amputation rate (53.8%) compared to the general surgery group (38.7%) ($p = 0.208$), suggesting that delayed or less aggressive debridement in orthopedic-managed patients may have contributed to a greater need for limb removal.

Limitations:

1. The sample size was relatively small, which may have affected the statistical power of some comparisons.
2. This study was conducted at a single center, limiting the generalizability of the findings to other healthcare settings.

Conclusion

Our findings suggest that surgical specialty significantly influences the outcomes of necrotizing fasciitis patients, with management by general surgeons being associated with better survival rates than those managed by orthopedic surgeons. These results highlight the need for further investigation into the factors contributing to these differences and emphasize the importance of multidisciplinary collaboration to improve patient outcomes. Future efforts should focus on addressing specialty-specific challenges and developing tailored interventions to optimize survival and overall care.

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